Computer Science - K-12 Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression a < c || a < b && !(a == c) evaluates to true?
   (a) a < c
   (b) a < b
   (c) a > b
   (d) a == b
2. Which is a valid expression in Python 3.5?
   (a) sort('ab')
   (b) sorted('ab')
   (c) "ab".sort()
   (d) 1/0
3. Which types of functions grow the slowest?
   (a) O(N^(1/2))
   (b) O(N^(1/4))
   (c) O(N^(1/N))
   (d) O(N)
4. A large Java program was tested extensively, and no errors were found. What can be concluded?
   (a) All of the preconditions in the program are correct.
   (b) All of the postconditions in the program are correct.
   (c) The program may have bugs.
   (d) Every method in the program may safely be used in other programs.
5. Which is the largest asymptotically?
   (a) O(1)
   (b) O(n)
   (c) O(n^2)
   (d) O(log n)

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the output of print(list[1:3]) for list = ['abcd', 786, 2.23, 'john', 70.2] is [786, 2.23].
7. True or False: The output of 'abc'[::-1] in Python 3 is 'abc', indicating that the string remains unchanged.
8. True or False: A function that averages numeric values in a column or row is the best way to detect data entry errors in a dataset.
9. True or False: In Python 3, the function float(x) is used to convert a string to an integer.
10. True or False: In Python3, the minimum value of the list l = [1,2,3,4] is 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the importance of clear specifications in programming. How would you modify the problem of finding an index of a value larger than a given item to address multiple valid outcomes?
12. Discuss the operator used for performing exponential calculations in Python 3. Explain its syntax, usage in code examples, and its significance in mathematical operations within programming.

End of Paper